HEALTH REVIEW / SC01

# One episode. A precise review request.

CHNAI LAB | 20 October 2026 | v0.1
Proposed research demonstration. Review not yet performed.

**Please assess measurement validity, wording and the permitted support action.** We are not asking you to approve a complete medical system. Suggested review time: 20-30 minutes; extend it if the evidence needs more work.

## The proposed demonstration

A simulated crew member has a pre-mission aerobic-capacity assessment and a comparable in-mission result. The software checks applicability, shows a source-linked comparison, supports an assigned onboard review and retains an optional Earth handoff while communication is unavailable.

| SIMULATED INPUTS | COMPUTED COMPARISON |
|---|---|
| Baseline 50.0 mL/min/kg<br>Follow-up 39.0 mL/min/kg | 39.0 / 50.0 = 78% retained<br>Below the 80% relative criterion |

## What comes from NASA

NASA-STD-3001 Volume 1, requirement 4003, sets an in-mission target of at least 80% of pre-mission aerobic capacity. Requirements 4001 and 4002 add separate mission-dependent EVA criteria. Passing one comparison does not establish overall fitness or EVA clearance. [N01]

Mission programs define their own health operations and technical requirements. This team has no real mission care plan. The proposed workflow and example values are CHNAI design choices. [N02]

## Wording to review

“This simulated assessment is 78% of the selected pre-mission baseline, below the 80% relative criterion. Confirm that the assessments are comparable. An assigned onboard review can be requested; this result does not establish a diagnosis or clearance.”

**Proposed support action:** Request onboard review. The request records an owner and an unresolved follow-up; it is not a treatment, emergency triage decision or proof that care occurred. Other health domains stay **Not assessed**.

REVIEW DECISIONS

# What we need you to resolve

For each item: Accept for this research demonstration / Change required / Cannot assess. Record the reason and exact replacement wording or policy.

| ID | Decision needed |
|---|---|
| HR01-02 | Quantity and comparability: what distinguishes a valid direct/indirect capacity assessment from ordinary exercise VO2? Specify baseline selection, protocol, units and relevant conditions. |
| HR03 | Applicability: confirm the intended use of 4003. Keep mission-dependent EVA criteria, overall health and medical clearance separate. |
| HR04-05 | Uncertainty and language: review the 78% explanation and the boundary cases on page 3. Remove any implied diagnosis, severity or biological certainty. |
| HR06 | Authority and action: is requesting review a suitable support step? Identify what a crew medical officer, other crew member and ground clinician may do in this hypothetical scenario. Do not assume an onboard role is a physician. |
| HR07 | Timing and unavailable support: identify any needed assessment-age or follow-up policy. If no policy exists, specify how to show it as unresolved rather than inventing a deadline. |
| HR08-09 | Handoff and disclosure: check the minimum necessary packet and the distinction between queued, received and medically reviewed. Confirm that all crew values are simulated and other health domains unassessed. |
| HR10 | Coverage: identify missing boundary cases or source conflicts that would make this demonstration misleading. |

## How to return the review

Use HEALTH_REVIEW_RESPONSE.md in the accompanying package. Record your relevant expertise and limits, exact brief version, findings, required changes and overall disposition. Attach supporting references for changes to the evidence interpretation.

A health or life-science student can provide a preliminary review within their expertise. A review of a hackathon demonstration is not clinical validation, NASA endorsement or authorization to treat. If an item exceeds your expertise, mark Cannot assess.

BOUNDARIES AND DISPOSITION

# Try to break the interpretation

All numerical cases below are invented test examples. They are neither NASA observations nor recommended physiological values. Proposed behavior still needs review.

| Case | Proposed behavior |
|---|---|
| B=50; F=40.0 | 80%: relative criterion met; no global healthy or EVA-safe label. |
| B=50; F=39.995 | 79.99%: below, even if a badge rounds to 80%. |
| B=50; F=40.005 | 80.01%: meets this comparison; retain original precision. |
| Baseline missing or zero | Not evaluable; no ratio, substituted baseline or reassuring fallback. |
| L/min versus mL/min/kg | Require reviewed conversion/context. Never guess body mass. |
| Method equivalence unknown | Hold the comparison for review; do not equate protocols automatically. |
| Uncertainty spans 80% | Qualify the point result; no invented probability or definitive biological classification. |
| Ordinary exercise VO2 | Do not relabel it as VO2max or a qualified capacity assessment. |
| Packet queued; no reviewer | Keep unresolved. Sending or receiving data is not delivered care. |

## What closes this review

A dated, scoped response; the exact evidence/version reviewed; a disposition; resolution of required changes; and team acknowledgment. Until those exist, G03 remains **Unreviewed**. Software tests and actual Mac performance also remain unexecuted.

## Primary reading and baseline

[N01] NASA: Human Performance, sections 4.1.1-4.1.3
[N02] NASA: Medical Operations, requirements 6006, 6008 and 6009
[I02] CHNAI workflow v0.1: SC01 and WQ01-WQ12

Official NASA pages rechecked 20 October 2026. Retained NASA-STD-3001 V1 Rev C is the project evidence baseline; a dated, reviewed release must govern any implementation. Keep this packet as preparation pending the event-rule clarification.